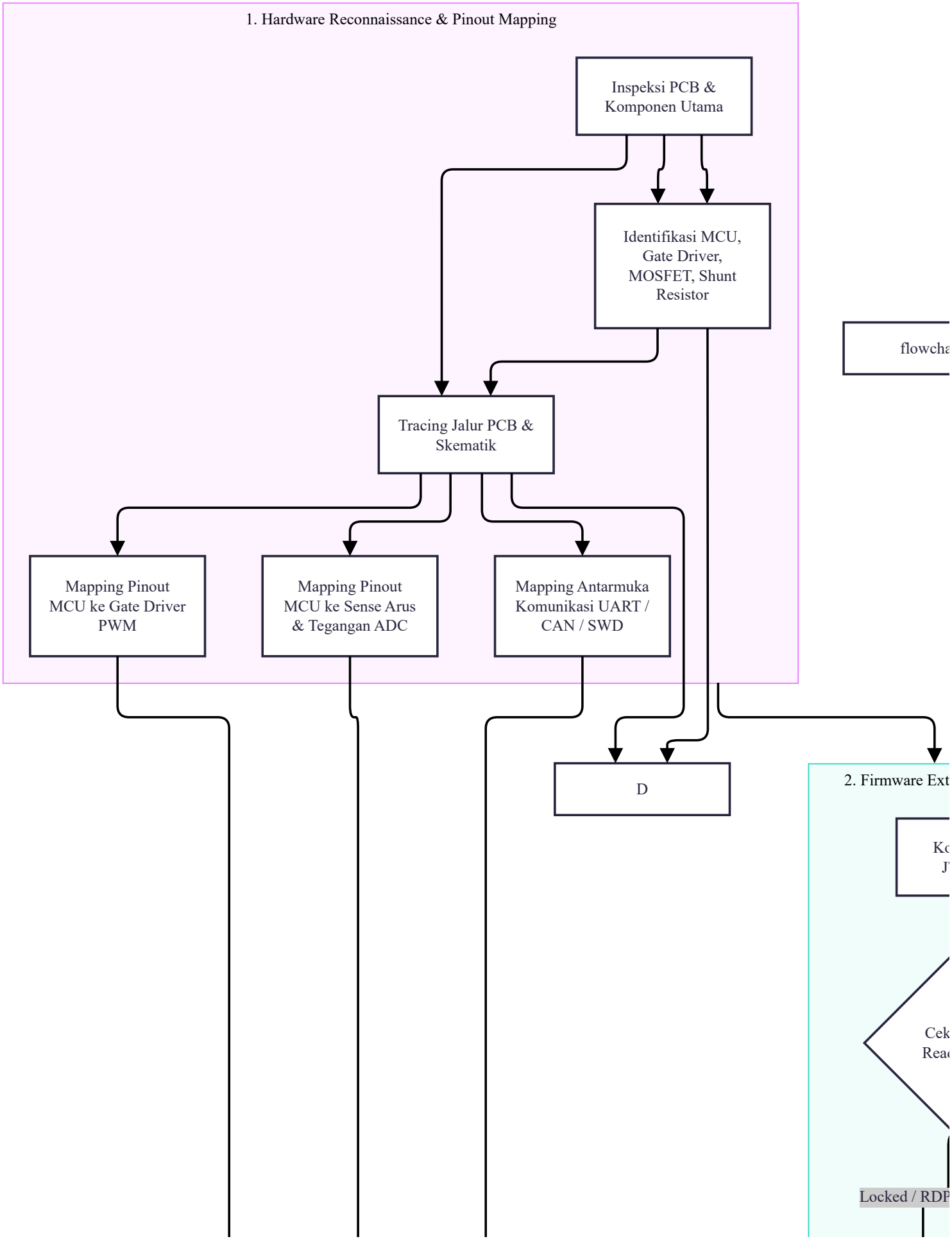
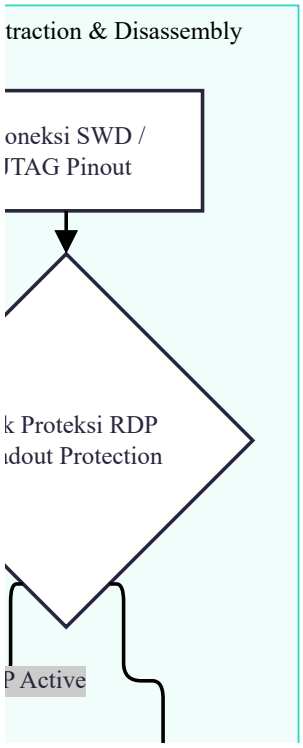
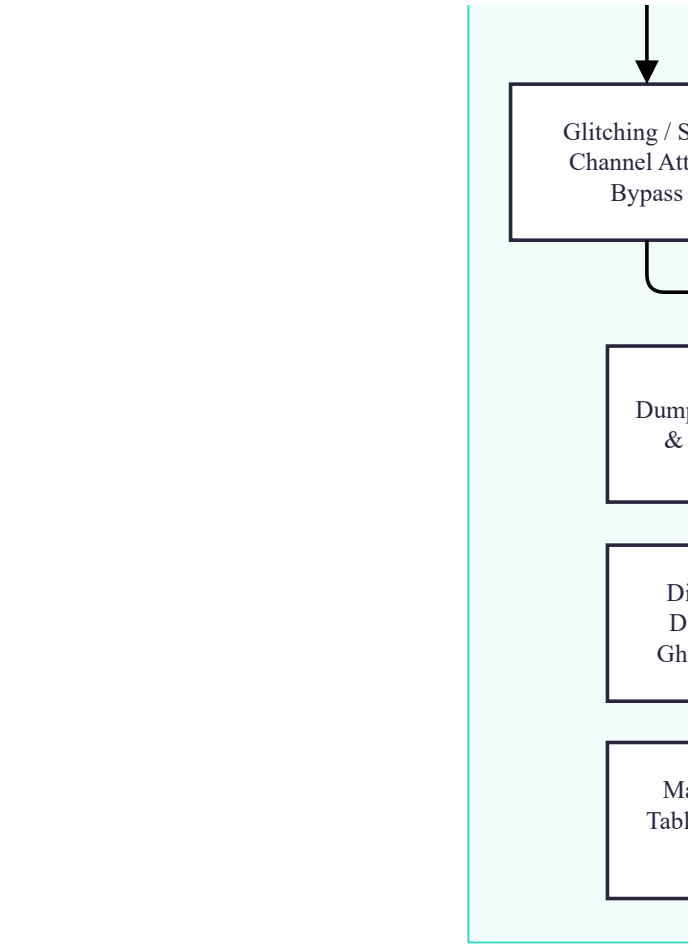
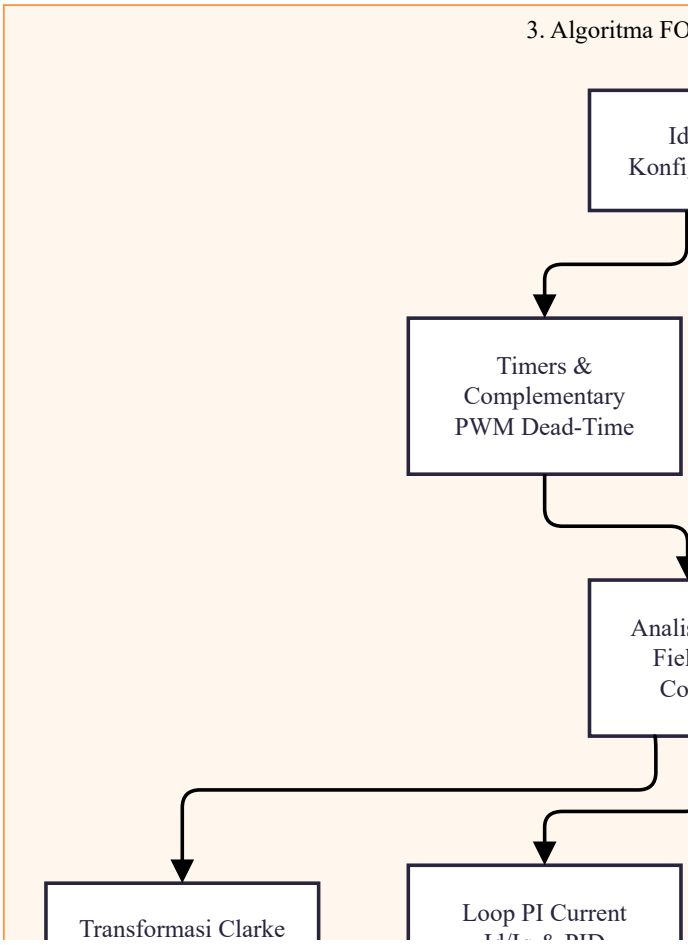
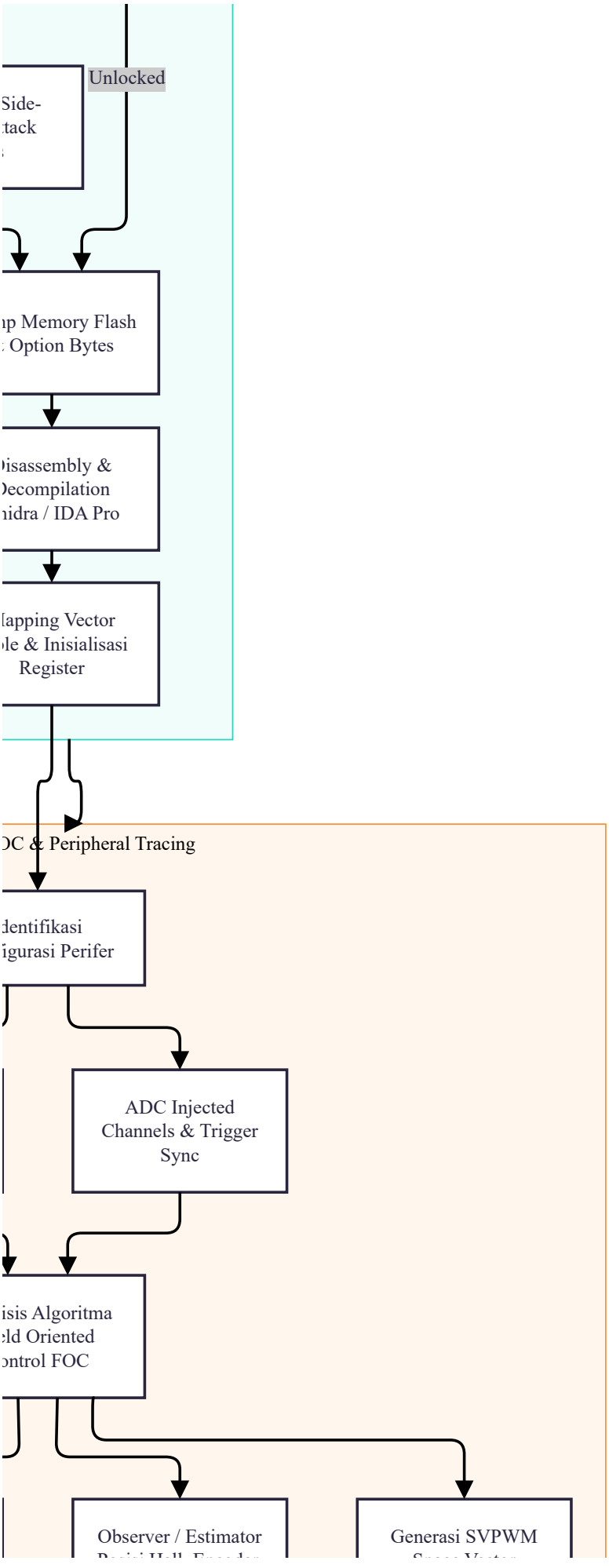

1. Hardware Reconnaissance & Pinout Mapping
Inspeksi PCB & Komponen Utama
Identifikasi MCU, Gate Driver, MOSFET, Shunt Resistor
Tracing Jalur PCB & Skematik
Mapping Pinout MCU ke Gate Driver PWM
Mapping Pinout MCU ke Sense Arus & Tegangan ADC
Mapping Antarmuka Komunikasi UART / CAN / SWD
D
flowcha
2. Firmware Ext
Cek
Rea
Locked / RDP

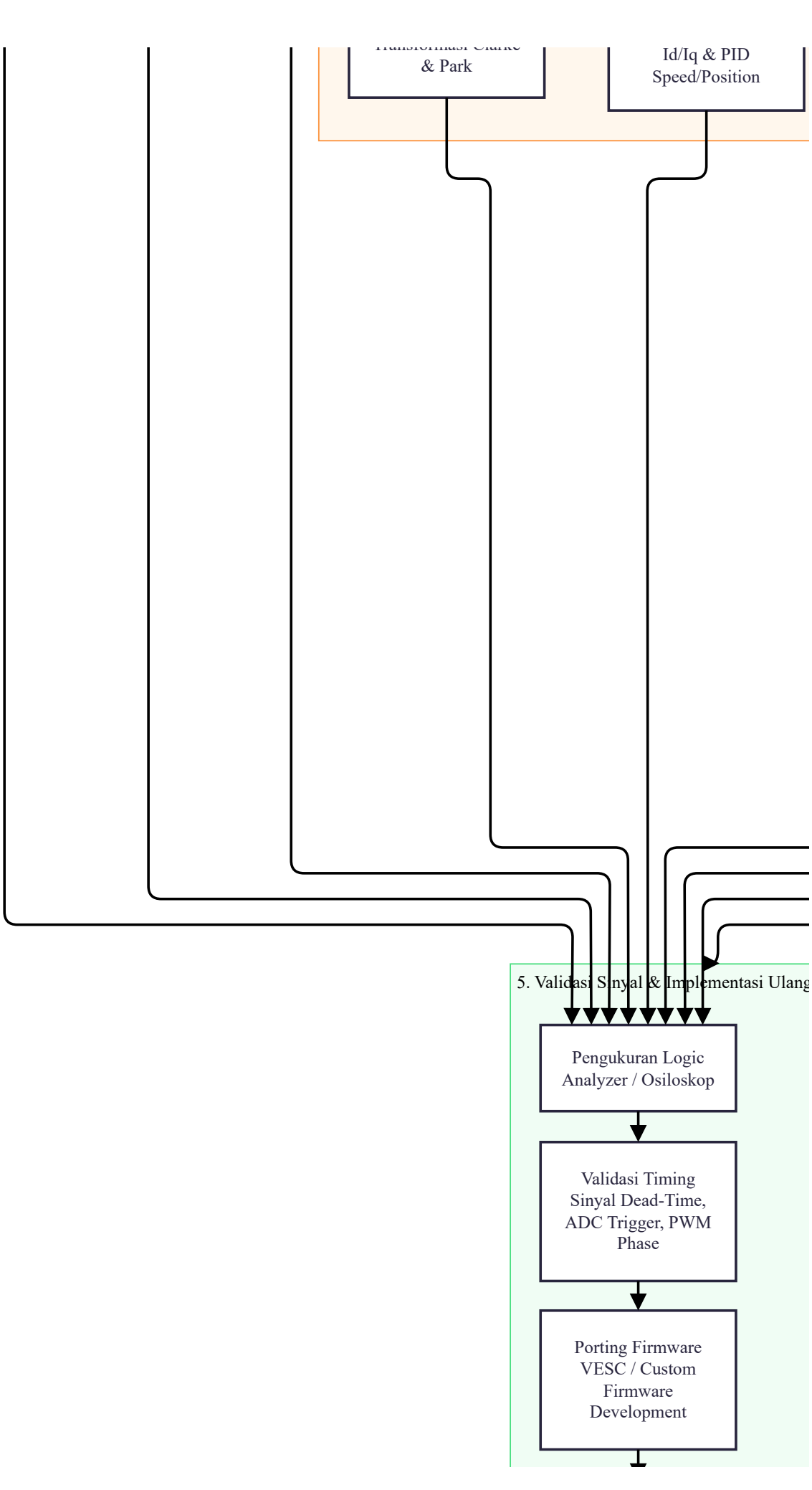

art
TD

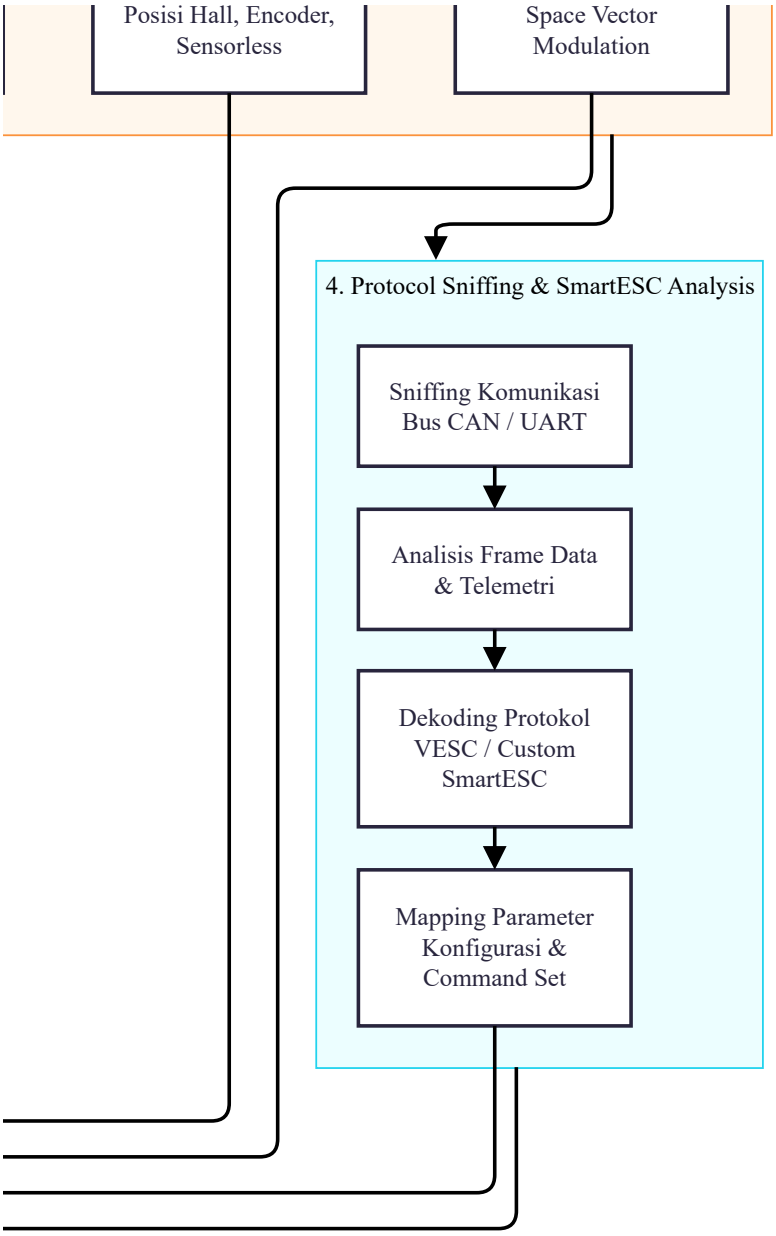

traction & Disassembly
oneksi SWD /
TAG Pinout
k Proteksi RDP
dout Protection
P Active


Glitching / S
Channel Att
Bypass
Dum
&
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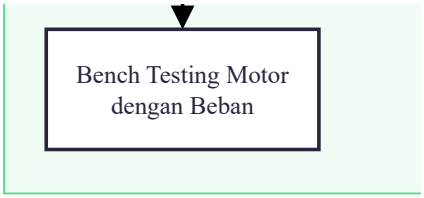

3. Algoritma FO
Id
Konfi
Timers &
Complementary
PWM Dead-Time
Anali
Fiel
Co
Transformasi Clarke
Loop PI Current


Unlocked
Side-
tack
ıp Memory Flash
Option Bytes
)isassembly &
)ecompilation
ıidra / IDA Pro
lapping Vector
le & Inisialisasi
Register
)C & Peripheral Tracing
dentifikasi
igurasi Perifer
ADC Injected
Channels & Trigger
Sync
isis Algoritma
ld Oriented
ontrol FOC
Observer / Estimator
Generasi SVPWM


& Park
Id/Iq & PID
Speed/Position
5. Validasi Sinyal & Implementasi Ulang
Pengukuran Logic Analyzer / Osiloskop
Validasi Timing Sinyal Dead-Time, ADC Trigger, PWM Phase
Porting Firmware VESC / Custom Firmware Development


Posisi Hall, Encoder, Sensorless
Space Vector Modulation
4. Protocol Sniffing & SmartESC Analysis
Sniffing Komunikasi Bus CAN / UART
Analisis Frame Data & Telemetri
Dekoding Protokol VESC / Custom SmartESC
Mapping Parameter Konfigurasi & Command Set

g


Bench Testing Motor
dengan Beban